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HARNES: Heuristic Algorithm for Routing and Network Efficiency Using SDN in Intelligent Transportation Systems

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ABSTRACT In this paper, we introduce *HARNES*, a heuristic and Software-Defined Networking (SDN)-enabled routing framework for dynamic vehicular networks. The framework enhances classical search algorithms by developing cluster-based extensions of *Theta** and *D**-Lite, which minimize computational overhead through localized decision-making and line-of-sight (LOS) optimization. By deploying *SDN* controllers at roadside units (RSUs), *HARNES* enables centralized coordination among clusters, thereby improving adaptability and scalability in large-scale networks. The proposed approach is evaluated against EG-Dijkstra (Evolving-Graph Dijkstra) and *A** on grid-based vehicular topologies using six performance metrics: computation time, discovery messages, path length, expanded nodes, average degree of expanded nodes, and path optimality ratio. Simulation results demonstrate that the cluster-based *Theta** achieves up to 42% reduction in computation time, 88% fewer discovery messages, and 47% fewer expanded nodes, while maintaining near-optimal path quality within 3.1% of optimality. Additional comparison with the original non-clustered *Theta** and *D**-Lite further confirms the effectiveness of the proposed cluster-based design. These findings highlight the potential of *HARNES* as a low-latency, scalable, and SDN-compatible routing solution for real-time intelligent transportation systems.

INDEX TERMS Cluster-based routing, EG-Dijkstra, intelligent transportation systems, SDN, VANETs, *Theta**.

I. INTRODUCTION

THE Internet-of-Vehicles (*IoV*) is becoming an important part of next-generation intelligent transportation systems (*ITS*), enabling communication among vehicles, roadside units (*RSUs*), and network controllers [1]. However, rapid mobility and frequent topology changes make reliable, scalable, and low-latency routing a persistent challenge in vehicular networks [2]. In this context, Software-Defined Networking (*SDN*) has emerged as a promising approach due to its support for centralized control and global network awareness [3]. Building on this direction, this paper presents *HARNES* (Heuristic Algorithm for Routing and Network Efficiency using *SDN*), a routing framework for intelligent vehicular communication.

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A. BACKGROUND WORK

Numerous intelligent routing frameworks have been proposed in recent years to cope with the dynamic nature of the *IoV* [4]. One prominent effort is *RADAR*, which employs the heuristic *Theta** algorithm within a tri-layered *SDN* architecture [5]. The first layer is responsible for real-time data acquisition from vehicles, capturing vital attributes such as position, velocity, direction, and identity. This data facilitates context-aware decision-making and supports interaction between vehicles and roadside infrastructure. The second layer uses an *SDN* edge controller to perform rapid intra-zone path discovery using *Theta**, while the third layer introduces a global controller that governs inter-zone routing,

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offering extended spatial coverage and improved fault tolerance [5].

The cluster-based VANET-Oriented Evolving Graph ($CVoEG$) model [6] evolved from the foundational VoEG framework [7] and was designed to enhance link reliability and route longevity in $VANETs$. It introduced selection criteria based on link stability for forming cluster members and cluster heads and applied spectral heuristics (Eigengap) to determine the number of clusters. This model further inspired the development of $CEG\text{-}RAODV$ (Clustered Evolving-Graph Reliable AODV), an extension of the Ad hoc On-Demand Distance Vector (AODV) routing protocol designed to discover the Most Reliable Journey (MRJ) from source to destination [6]. In this line of work, clustering and reliability-aware communication have played an important role in improving route stability in vehicular environments [8].

Another significant advancement is the SDN-enabled cluster-based evolving graph routing scheme ($SE\text{-}CEGRS$) [9]. $SE\text{-}CEGRS$ integrates the centralized management capability of SDN with the $CVoEG$ model. Unlike traditional $CVoEG$ [6], where both intra- and inter-cluster routing rely on a distributed EG-Dijkstra approach, $SE\text{-}CEGRS$ retains distributed routing within clusters while using a centralized SDN controller for inter-cluster communication. This hybrid design extends communication coverage and improves coordination across clustered vehicular regions.

In parallel, several heuristic-based approaches have been explored to reduce overhead and improve responsiveness [10]. For example, the heuristic path search and multi-attribute decision-making-based routing (HMDR) method combines heuristic path search with multi-attribute decision-making to enhance the transmission of safety messages in urban vehicular networks [11]. Another approach integrates K-means clustering with heuristic optimization to improve routing stability and reliability in $VANETs$ [12]. These studies demonstrate the potential of heuristic techniques to support efficient and adaptive routing under dynamic vehicular conditions.

Overall, prior studies show that both clustering-based and heuristic-based routing strategies offer meaningful advantages in vehicular environments. They also indicate the potential of combining localized heuristic efficiency with the broader coordination capability of SDN. Building on this direction, our proposed framework, $HARNES$, introduces cluster-based enhancements of $Theta^*$ and D^*-Lite within an SDN-coordinated architecture.

B. MOTIVATIONS AND LIMITATIONS

In dynamic vehicular networks, the main motivation is to achieve fast, scalable, and reliable route discovery under frequent topology changes, dense traffic conditions, and strict communication delays [4]. Although SDN provides centralized visibility and flexible control, efficient routing in clustered vehicular environments still requires low-overhead path computation that can quickly adapt to local changes [13]. This makes it necessary to combine the global coordination capability of SDN with routing strategies that can reduce unnecessary exploration and respond efficiently in real time.

Despite recent advancements in SDN-enabled and clustering-based routing schemes for vehicular networks, several challenges still exist. Frameworks such as $RADAR$ [5] and $SE\text{-}CEGRS$ [9] have demonstrated the value of integrating SDN with layered and cluster-based architectures to improve route management and adaptability. However, while $SE\text{-}CEGRS$ is effective at separating intra- and inter-cluster routing, it relies on non-heuristic algorithms, such as EG-Dijkstra, as part of its clustered design. Although EG-Dijkstra guarantees optimal path selection, its exhaustive search nature incurs high control overhead, especially as network density and the number of clusters increase. This becomes a bottleneck in real-time scenarios where fast route computation and communication efficiency are critical.

Moreover, non-heuristic routing approaches are less flexible in dynamic environments and often fail to prioritize promising routes early in the exploration process. This lack of guided search increases the number of expanded nodes and discovery messages, particularly in settings with frequent topology changes. In contrast, heuristic algorithms such as $Theta^*$ and D^*-Lite provide spatial awareness and can exploit geometric or topological features that significantly reduce routing overhead [10]. Therefore, there is a clear need to extend clustered SDN-based routing from non-heuristic path discovery toward a heuristic-aware design that can better balance efficiency and scalability.

Motivated by these observations, our work introduces a new framework, $HARNES$, that combines heuristic pathfinding with a cluster-based scheme under SDN coordination. By extending $SE\text{-}CEGRS$ into a heuristic-aware design, $HARNES$ leverages cluster-based $Theta^*$ and D^*-Lite algorithms to balance local efficiency with global visibility. This hybrid approach directly addresses the scalability, responsiveness, and control-overhead limitations of existing non-heuristic, cluster-based SDN routing schemes.

C. CONTRIBUTIONS

This paper introduces $HARNES$, a routing framework that combines heuristic pathfinding with SDN coordination to improve scalability and responsiveness in dynamic vehicular networks. The main contributions of this work are summarized as follows:

- We propose cluster-based extensions of $Theta^*$ and D^*-Lite for vehicular routing. These extensions reduce the search space by limiting route discovery to local clusters and using line-of-sight (LOS) shortcuts.
- We integrate the proposed routing scheme with an SDN architecture, where controllers deployed at $RSUs$ support inter-cluster coordination while local routing is handled within each cluster.
- We evaluate $HARNES$ in a grid-based vehicular environment with temporal mobility using six performance metrics: Computation Time (CT), Discovery Messages (DM), Path Length (PL), Expanded Nodes (EX), Average Degree of Expanded Nodes (AD), and Path Optimality Ratio (POR).

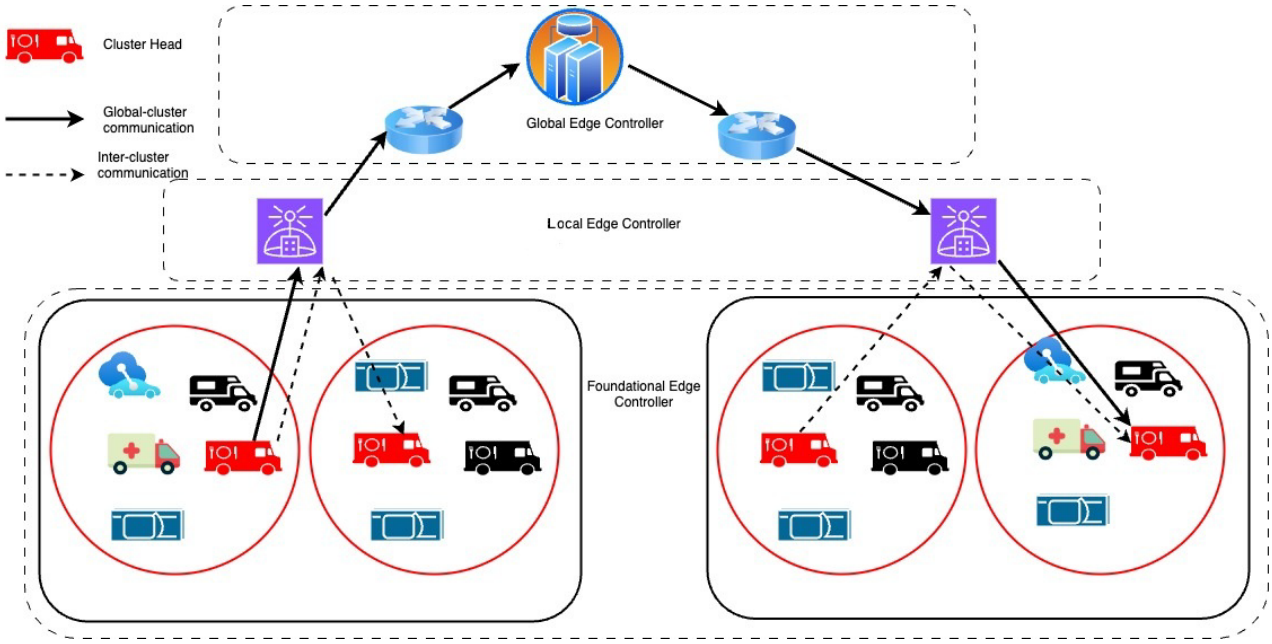


FIGURE 1. Hierarchical SDN-VANET architecture illustrating the integration of the vehicle layer with the Local Edge Controllers (*LECs*) and the Global Edge Controller (*GEC*), coordinated by the Foundational Edge Controller (*FEC*), for enhanced clustering and routing management.

- The results show that cluster-based $Theta^*$ performs best overall, achieving up to 42% lower computation time, 88% fewer discovery messages, and 47% fewer expanded nodes while maintaining near-optimal path quality. Additional comparison with the original non-clustered $Theta^*$ and D^*-Lite further confirms the effectiveness of the proposed cluster-based design, with more pronounced gains observed for D^*-Lite.

The remainder of this paper is organized as follows. Section II presents the system model for clustering and routing in integrated SDN-VANET environments. Section III describes the implementation details of the proposed $HARNES$ framework, focusing on heuristic routing dynamics within Software-Defined Vehicular Networks (SDVN). Section IV provides the performance evaluation and detailed discussion of results across multiple metrics. Finally, Section V concludes the paper and outlines directions for future research.

II. SYSTEM MODEL FOR CLUSTERING AND ROUTING IN INTEGRATED SDN-VANET

This section outlines the integration of SDN with the IoV to extend communication boundaries and enhance manageability, particularly in complex highway scenarios. Our model employs advanced clustering techniques alongside heuristic-based route discovery, utilizing D^*-Lite and $Theta^*$ algorithms. This strategy uses SDN to enable centralized control and scalable communication across wide highway networks. A summary of the symbols used throughout the system model and subsequent formulations is provided in Table 1. This table outlines the key elements and parameters integral to our $HARNES$ framework.

A. VEHICLE EQUIPMENT AND COMMUNICATION INFRASTRUCTURE

Each vehicle in the network is equipped with advanced onboard units (OBUs) and Global Positioning System (GPS) modules. The $OBUs$ enable robust communication, while the GPS modules provide precise geolocation data essential for dynamic topology mapping [14]. The network infrastructure is reinforced by 5G-enabled $RSUs$, which are crucial for high-throughput data exchange and real-time traffic management [15].

The SDN architecture forms the backbone of the network with a three-tier hierarchical controller as illustrated in Figure 1. The foundational layer, coordinated by the Foundational Edge Controller (FEC), consists of individual vehicles, each equipped with advanced communication technology. The intermediate layer is managed by Local Edge Controllers ($LECs$), which handle communication and data processing within their respective zones, supporting local decision-making. The Global Edge Controller (GEC) provides centralized oversight of the entire network. This controller ensures consistent rule enforcement and optimal route selection across all areas. The three-tiered architecture significantly boosts the network's scalability, manageability, and responsiveness [5].

B. COMMUNICATION MODEL

Our communication model is split into intra- and inter-cluster communication, using link reliability to optimize communication among vehicles within clusters. In the proposed hierarchy, foundational operations such as vehicle information exchange, cluster formation, and cluster-head selection are separated from the routing coordination process, which is performed according to the communication scope.

TABLE 1. List of symbols and their descriptions in the SDN-VANET system model.

Symbol	Description
V	Set of all vehicles
D^*	D^*-Lite algorithm
Θ^*	$Theta^*$ algorithm
BC	Betweenness Centrality for cluster head selection
$RSUs$	For data exchange
$R_l(t)$	Link reliability at time t
D_l	Active link duration
T_R	Relative transmission range
$\Delta\nu$	Differential velocity between vehicles
$g(T)$	Probability density function of communication duration
$\sigma_{\Delta\nu}$	Standard deviation of differential velocity
$\mu_{\Delta\nu}$	Mean of differential velocity
p	A path from node S to D
$\text{Paths}_{D^*}(S, D)$	Paths from S to D via D^*-Lite
$\text{Paths}_{\Theta^*}(S, D)$	Paths from S to D via $Theta^*$
v_i, v_j	Vehicles in a communication link
σ_{SD}	Number of shortest paths from S to D
$\sigma_{SD}(v)$	Paths through vehicle v
$\text{Cluster}(v)$	Vehicles in the cluster led by v

1) INTRA-CLUSTER COMMUNICATION

Within clusters, vehicles continuously exchange essential data, such as speed and position, via beacon messages directed to nearby *RSU* or designated cluster heads. This local communication enables the fast sharing of important traffic information within the cluster, supported by strong, reliable connections between nearby vehicles. After clusters are formed, intra-cluster routing coordination is handled at the local controller level, where the *LEC* supports efficient communication among vehicles belonging to the same cluster.

Cluster heads are strategically selected to optimize data transmission efficiency within clusters. The selection process utilizes Betweenness Centrality (*BC*), which identifies vehicles that act as central communication hubs within the network [16]. By selecting vehicles with significant *BC* values, the system ensures that data relay across the network is efficient and robust, enhancing the overall stability and connectivity within the cluster. This selection process highlights the critical role of strategic node placement for effective communication within clusters. The formation of clusters and the selection of cluster heads, guided by link reliability, are discussed in greater detail in the clustering model subsection.

2) LOCAL INTER-CLUSTER COMMUNICATION

For communications that extend beyond individual cluster boundaries but remain within the same zonal area managed by a particular *RSU*, coordination is still maintained locally. In this case, the *RSU* works in conjunction with the *LEC* to efficiently manage data transmission across clusters. When the target vehicle is located within another cluster under the same *RSU*, the originating *CH* sends the communication request directly to the *RSU* [6]. The *RSU* then promptly forwards this data to the destination cluster's *CH*. This localized handling of requests ensures rapid response times and minimizes the communication overhead across the network [17]. In this case, the *LEC* is responsible for local routing coordination within the same zonal domain, including intra-cluster support and inter-cluster communication among neighboring clusters. It also serves as the first point of escalation when the destination cannot be reached via direct intra-cluster communication.

3) GLOBAL INTER-CLUSTER COMMUNICATION

When a destination vehicle is not found within the local *RSU* routing table, the request is forwarded to the Global Edge Controller (*GEC*). This controller oversees network-wide communication, efficiently routing data across extensive geographic areas [18]. The *GEC* investigates the network topology to determine the most efficient path to the destination, aiming to select the most reliable route with minimal delay [19]. This centralized approach ensures optimal routing decisions, enhancing network scalability and integrity. This process distributes communication loads across the network. It prevents any single RSU from becoming overwhelmed by inter-zonal traffic and dynamically adjusts routes based on real-time network conditions and vehicle movements. In summary, *GEC* supports robust and adaptive connectivity throughout the *SDN-VANET* framework [5]. Thus, the division of responsibility is hierarchical: the *LEC* handles local routing operations within the same zonal area, whereas the *GEC* is invoked only when the destination lies outside the local controller's view and inter-zonal coordination is required.

C. LINK RELIABILITY MODEL

Link reliability in vehicular networks is defined as the statistical probability that a direct communication link between two vehicles, v_i and v_j, remains continuously available over a specified time interval t. This measure can be affected by various factors, including signal strength, interference, vehicle density, speed, distance, and physical obstacles [2]. The robustness of these links is important for stable and reliable communications among vehicles in both inter- and intra-cluster setups.

The reliability $R_l(t)$ of a communication link l is mathematically represented as:

$$R_l(t) = \int_t^{t+D_l} g(T)\, dT \qquad \text{if} \quad D_l > 0$$

where D_l is the duration for which the link is available, dictated by the relative transmission range T_R and the differential velocity $\Delta\nu$ between the two vehicles:

$$D_l = \frac{T_R}{\Delta\nu}$$

The probability density function (pdf) $g(T)$ of the communication duration T takes the form:

$$g(T) = \frac{T_R}{\sigma_{\Delta\nu}\sqrt{2\pi}T^2} e^{-\frac{(T_R/T-\mu_{\Delta\nu})^2}{2\sigma^2_{\Delta\nu}}}$$

The aforementioned reliability metric is used in our clustering model. The purpose of said metric is to ensure that cluster heads and their members are connected through highly reliable communication links. This setup not only supports effective intra-cluster communication but also enhances the efficiency of inter-cluster data transmission, which is crucial in dynamic vehicular environments [20]. By utilizing link reliability in cluster formation and maintenance, our model aims to optimize network stability and throughput.

D. CLUSTERING MODEL

Our clustering model utilizes Betweenness Centrality to dynamically select cluster heads and coordinate cluster formations. This metric enhances network stability and efficiency by ensuring that cluster heads are strategically positioned to facilitate effective data transfer [21]. The reliability of the link forms the basis for calculating the shortest path in our model, thereby affecting the calculation of BC by influencing the path selection process [22]. We employ D^*-Lite and $Theta^*$ separately to dynamically compute the most reliable paths, the set of reliable routes through D^*-Lite:

$$\text{Reliable Path}_{D^*}(S, D) = \max_{p \in \text{Paths}_{D^*}(S,D)} \left\{ \sum_{(v_i, v_j) \in p} R_l(t_{ij}) \right\} \quad (1)$$

In case of $Theta^*$, the reliable paths are:

$$\text{Reliable Path}_{\Theta^*}(S, D) = \max_{p \in \text{Paths}_{\Theta^*}(S,D)} \left\{ \sum_{(v_i, v_j) \in p} R_l(t_{ij}) \right\} \quad (2)$$

where p denotes a path in the set of all possible paths from S to D. $R_l(t_{ij})$ represents the link reliability between vehicles v_i and v_j at time t_{ij}. $\text{Paths}_{D^*}(S, D)$ and $\text{Paths}_{\Theta^*}(S, D)$ denote all paths discovered by D^*-Lite and $Theta^*$ respectively.

1) CLUSTER HEAD SELECTION

Cluster heads are selected based on their high BC values, indicating their important role within the network's communication framework. Vehicles that consistently demonstrate the highest BC are chosen as cluster heads:

$$BC(v) = \sum_{\substack{s,t \in V \\ s \neq t,\ v \notin \{s,t\}}} \frac{\sigma_{st}(v)}{\sigma_{st}} \quad (3)$$

where σ_{st} is the number of reliability-weighted shortest paths from node s to node t, computed using the paths identified by either D^*-Lite or $Theta^*$. The $\sigma_{st}(v)$ denotes the paths among σ_{st} that traverse through vehicle v. This criterion ensures that the cluster head is not only central but also on the most stable and reliable paths, enhancing the efficiency of intra- and inter-cluster communication.

2) CLUSTER FORMATION

After identifying the group heads, the vehicles are grouped around these heads. This approach is enhanced by considering vehicle mobility patterns and communication requirements. Consequently, cluster formation remains flexible and adaptable to the network's dynamic characteristics [23]. The clusters aim to reduce total communication delays and enhance data transfer rates, supported by the robust link reliability guaranteed by the path calculations:

$$\text{Cluster}(v) = \{u \in V | \text{path reliability to } v \text{ is maximized}\} \quad (4)$$

Here, $\text{Cluster}(v)$ denotes the set of vehicles combined in clusters led by head v. It ensures that every member vehicle maintains a reliable communication link to the cluster head, contributing to a more reliable and efficient vehicular network [24].

Although clustering improves routing scalability by narrowing the search space, it introduces additional coordination overhead during cluster formation and cluster-head selection. This overhead is incurred before or alongside route discovery and may vary depending on network density and the selected cluster heads. In the current framework, this trade-off is acceptable because the routing phase benefits from reduced computation time, fewer discovery messages, and lower search effort. Nevertheless, the impact of re-clustering frequency and cluster-head sensitivity under highly dynamic mobility remains an important aspect for further investigation.

E. ROUTING MODEL

The proposed *HARNES* framework integrates SDN with clustering and heuristic route discovery algorithms. This integration leverages their combined strengths to enable efficient and adaptive routing in VANETs. Our proposed routing framework uses the D^*-Lite [25], [26] and $Theta^*$ [5], [27] algorithms, which are tailored for dynamic, efficient pathfinding under real-time traffic conditions, ensuring optimal communication flow and enhanced network resilience across varied vehicular environments [28].

1) D^*-LITE ALGORITHM

D^* -Lite, originally proposed by Koenig and Likhachev [25], is pivotal for its efficient recalculation of routes in response to changes in the network topology, making it especially suitable for environments with frequent dynamic changes [26], [29]. It benefits from the clustering model by focusing route recalculations within and between clusters. As a result, it significantly reduces overhead and improves response time as well:

$$\text{Update}(s) = \min_{s' \in \text{Succ}(s)} (c(s, s') + g(s')) \quad (5)$$

where $c(s, s')$ represents the cost from node s to s', and $g(s')$ is the cost from s' to the goal. This localized updating mechanism uses cluster-head information to optimize route discovery.

2) $THETA^*$ ALGORITHM

$Theta^*$, originally introduced by Nash et al. [27], extends the capabilities of A^* by allowing a more direct routing approach. It uses *LOS* operations between non-adjacent nodes, which subsequently reduces overall travel times and enhances

network efficiency by potentially bypassing intermediate nodes [5], [30]:

$$\text{LOS}(s, s') = \begin{cases} \text{true}, & \text{if no obstacles between } s \text{ and } s', \\ \text{false}, & \text{otherwise.} \end{cases} \tag{6}$$

In our *HARNES* framework, the *Theta** algorithm takes advantage of real-time traffic and topology data provided by Edge controllers. As a result, it ensures optimal routing in highly dynamic networks.

III. IMPLEMENTING HARNES: HEURISTIC CLUSTER-BASED ROUTING IN SOFTWARE-DEFINED VEHICULAR NETWORKS (SDVN)

A. HARNES OVERVIEW

HARNES enhances routing in *SDVN* by combining a cluster-based model with heuristic, *LOS*-aware route discovery. The flow diagram in Figure 2 illustrates how *HARNES* works step by step. It shows the progression from vehicle information exchange and network graph formation to cluster formation and CH selection, followed by hierarchical route discovery through intra-cluster, local inter-cluster, and global inter-cluster communication levels. In this hierarchy, the *LEC* is responsible for local routing coordination within the same zonal domain, including intra-cluster and local inter-cluster communication, while the *GEC* is responsible for routing requests that must be escalated across multiple zones when the destination is outside the local controller's view. Here is how *HARNES* enhances routing dynamics across different communication levels within vehicular networks:

- **Foundational Edge Controller (FEC):** At this level, *HARNES* focuses on foundational operations such as vehicle information exchange, topology initialization, cluster formation, and CH selection [31].
- **Local Edge Controller (*LEC*):** The *LEC* addresses local routing operations, including intra-cluster and inter-cluster communications managed within the same zonal area. It serves as the first routing escalation level once the destination is not reachable through direct intra-cluster communication. It involves dynamic routing adjustments to enhance data transfer between clusters based on current network conditions and link reliability, facilitating swift and effective regional communication [32].
- **Global Edge Controller (*GEC*):** When cluster interactions span across different regional zones, *HARNES* engages the *GEC* to manage extensive network communications. This controller oversees data routing across the various *LEC*s, applying heuristic algorithms to maximize network efficiency and ensure uniform service quality over large areas [33]. Thus, the *GEC* is invoked only when the destination lies outside the local zonal domain or cannot be resolved at the *LEC* level.

HARNES integrates these mechanisms into the *SDN* framework to effectively address the complexities and dynamic nature of vehicular networks. By combining advanced clustering with dynamic heuristic routing, *HARNES* aims to significantly enhance communication performance and reliability across vehicular networks, facilitating more advanced *ITS* applications and services.

B. ROUTE DISCOVERY IN INTRA-CLUSTER COMMUNICATION FOR HARNES

This section details the intra-cluster route discovery mechanisms used in *HARNES*, which leverage both the D^*-Lite and Θ^* algorithms. These methods effectively address the dynamic challenges of finding routes within clusters structured under the *HARNES* framework.

1) ALGORITHM 1: D^*-LITE FOR INTRA-CLUSTER DYNAMIC PATHFINDING

D^* -Lite is adept at adapting to dynamic changes within a cluster, such as traffic variations or obstacles, making it ideal for environments that experience frequent change.

- **Initialization:** Set $g(u)$ and $\text{rhs}(u)$ for all nodes u to infinity, except for the destination D, where $\text{rhs}(D) = 0$. This setup initiates the backward search from the destination.
- **Priority Queue Management:** Manage nodes in a priority queue, ordered by a key combining their cost metrics and heuristic estimates, to efficiently manage path updates.
- **Path Optimization Loop:** Continuously update paths by extracting the node with the lowest key, adjusting nodes and their neighbors' path costs based on newly discovered shorter paths.
- **Neighbor Update:** If a shorter path through a node is found, adjust the rhs values of its neighbors and reflect these changes in the priority queue.
- **Path Reconstruction:** After achieving path cost consistency at the destination ($g(D) = \text{rhs}(D)$), backtrack from the destination to the source using predecessor links to trace the optimal path.

2) ALGORITHM 2: Θ^* FOR INTRA-CLUSTER ANY-ANGLE PATHS

Θ^* optimizes path length by enabling direct paths through line-of-sight checks, which is suitable for stable cluster conditions.

- **Initialization:** Start with the source node S, setting $g(S) = 0$ and placing S into an open list sorted by the total estimated cost $(g + h)$.
- **Exploration Loop:** Process nodes from the open list, exploring direct paths to neighbors using line-of-sight from the current node's parent, and updating path costs where direct routes are feasible.
- **Line-of-Sight Optimization:** If a direct line-of-sight path from S's parent to a neighbor v offers a lower cost, update the path cost to v, adjusting v's parent to the direct predecessor if applicable.
- **Path Updating:** Continuously refine the path to reflect the shortest and most direct routes, leveraging the geometric freedom provided by Θ^*.
- **Path Completion:** Conclude the route discovery once the destination D is reached, tracing back the path from D to S via parent links to reconstruct the route.

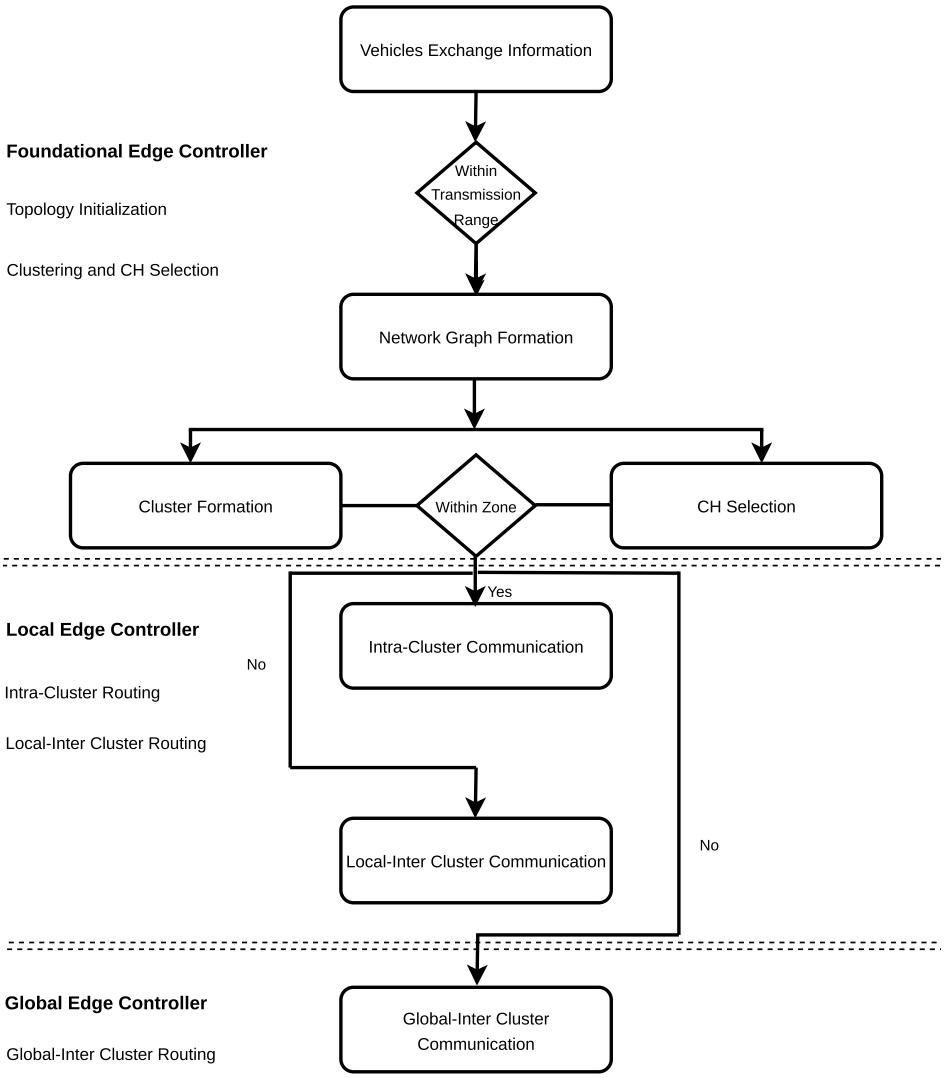


FIGURE 2. Flowchart of the HARNES protocol in SDVN, from topology initialization and clustering to hierarchical route discovery across the *FEC*, *LEC*, and *GEC* communication levels.

These algorithms collectively enhance intra-cluster communication efficiency in *HARNES*, providing robust, efficient route management tailored to the dynamics of vehicular network environments.

C. LEC COMMUNICATION: INTER-CLUSTER COORDINATION

This subsection details the inter-cluster communication mechanisms coordinated by $LECs$ within the *HARNES* framework, which utilize both the D^*-Lite and $Theta^*$ algorithms to ensure efficient data transmission across cluster boundaries.

1) ALGORITHM 3: D^*-LITE FOR LOCAL INTER-CLUSTER COORDINATION

D^* -Lite is adapted to dynamically manage route updates effectively when network topology changes occur within the local RSU's range. This approach is ideal for scenarios where inter-cluster communication needs frequent updates due to variables such as vehicle mobility or temporary obstructions.

- **Initialization:** Initialize all nodes u with $g(u) = \infty$ and $rhs(u) = \infty$, except for the source, where $rhs(source) = 0$, to start the pathfinding from the source cluster head.
- **Priority Queue Management:** Manage nodes in a priority queue based on a key that combines their cost metrics and heuristic estimates, facilitating efficient path updates.
- **Inter-Cluster Pathfinding Loop:** Continuously update paths by prioritizing nodes with the lowest keys and adjusting the paths of neighboring clusters as necessary.
- **Path Updating:** Adjust the rhs values of neighboring clusters and update their positions in the priority queue to reflect changes in path costs.
- **Path Reconstruction:** Once the destination cluster head is reached, reconstruct the inter-cluster path using predecessor relationships to ensure the most efficient routing.

Algorithm 1 D^*-Lite for Intra-Cluster Communication

Require: Source vehicle S, Destination vehicle D, Cluster topology
Ensure: Optimal path from S to D

Initialize all nodes u: $g(u) = \infty$, $rhs(u) = \infty$
Set $rhs(D) = 0$ (goal initialization)
Insert D into priority queue U with keys calculated by:
  $Key(u) = [\min(g(u), rhs(u)) + h(S, u), \min(g(u), rhs(u))]$
while U is not empty **do**
  $u = U$.pop()
  if $g(u) > rhs(u)$ **then**
    $g(u) = rhs(u)$
    for each neighbor v of u **do**
      $rhs(v) = \min(rhs(v), c(v, u) + g(u))$
      U.insert(v, calculate key)
    end for
  else
    Set $g(u) = \infty$
    for each neighbor v **do**
      Update $rhs(v)$
      U.insert(v, calculate key)
    end for
  end if
end while
Construct path from S to D using predecessors from D

Algorithm 2 Θ^* for Intra-Cluster Communication

Require: Source vehicle S, Destination vehicle D, Cluster map
Ensure: Shortest path from S to D using LOS checks

Initialize: $g(S) = 0$, $parent(S) = \text{None}$
Open list Q prioritized by $f = g + h$
Insert S into Q
while $Q \neq \emptyset$ **do**
  $u = Q$.pop()
  if $u = D$ **then break**
  end if
  for each neighbor v of u **do**
    if $LOS(\text{parent}(u), v)$ and $g(\text{parent}(u)) + c(\text{parent}(u), v) < g(v)$ **then**
      $g(v) = g(\text{parent}(u)) + c(\text{parent}(u), v)$
      $parent(v) = \text{parent}(u)$
      Insert/update v in Q
    end if
  end for
end while
Reconstruct path from S to D via parent pointers

2) ALGORITHM 4: *THETA** FOR ENHANCED PATHFINDING BETWEEN CLUSTERS

Theta* optimizes pathfinding by enabling direct paths between cluster heads when line-of-sight is possible, optimizing route length, and reducing communication delays across clusters.

- **Initialization:** Begin with the source cluster head, setting $g(source) = 0$ and placing it in an open list sorted by total cost $(g + h)$.
- **Direct Pathfinding Loop:** Explore direct connections through line-of-sight checks from the current node's parent, updating path costs for more direct routes when feasible.
- **Line-of-Sight (LOS) Assessment:** If a direct LOS from the source cluster head's parent to a neighbor cluster head offers a lower cost path, update the path cost and adjust the parent accordingly.
- **Continuous Path Optimization:** Dynamically refine paths by reassessing direct routes and updating route costs to reflect the most efficient paths between clusters.
- **Completion of Route Discovery:** Finalize the routing process once the destination cluster head is effectively connected, tracing back from the destination to the source to outline the full inter-cluster route.

D. GLOBAL EDGE CONTROLLER COMMUNICATION: INTER-CLUSTER COMMUNICATION

This subsection explores the pivotal role of the global SDN controller in facilitating efficient inter-cluster communication within the $HARNES$ framework. The global SDN controller employs the advanced routing algorithms D^*-Lite and *Theta** to effectively manage communication across the boundaries of extensive vehicular networks.

1) ALGORITHM 5: D^*-LITE FOR GLOBAL INTER-CLUSTER COMMUNICATION

D^* -Lite is instrumental in managing global path updates that are crucial for establishing robust communication links between clusters, especially those beyond the local RSU operational range.

- **Initialization:** Set $g(u)$ and $rhs(u)$ for all nodes u to infinity, except for the source, where $rhs(source) = 0$. This setup initiates a backward search from the source node, adapting to the global context.
- **Priority Queue Management:** Utilize a priority queue to efficiently manage nodes based on their cost metrics and heuristic estimates, which are crucial for recalculating paths on a global scale.
- **Global Pathfinding Loop:** Continuously update paths by processing nodes with the lowest keys, recalculating the paths of nodes across distant clusters.
- **Path Updating:** Adjust the rhs values of neighboring nodes based on changes in path costs, ensuring that the queue reflects the most current path information.
- **Path Reconstruction:** Once the destination node's path costs stabilize, reconstruct the path by tracing predecessors from the destination back to the source to ensure the optimal route is followed.

2) ALGORITHM 6: *THETA** FOR EXTENDED PATH OPTIMIZATION

Theta* leverages direct line-of-sight paths to enhance communication efficiency across distant clusters, optimizing the global network structure.

- **Initialization:** Start with the source node, setting $g(source) = 0$ and placing it into an open list sorted by the total estimated cost ($g + h$).
- **Line-of-Sight Pathfinding Loop:** Extend paths directly using line-of-sight checks from a node's parent, adjusting path costs for nodes where direct lines are viable.
- **Direct Path Assessment:** Apply direct paths between distant cluster heads when feasible to reduce cumulative travel distances and streamline communication paths.
- **Path Updating:** Regularly refine path costs to reflect shorter or more direct routes, thus enhancing overall routing efficiency.
- **Route Completion:** End the route discovery process once the destination is reached by retracing steps from the destination back to the source to compile the most efficient route.

IV. RESULTS AND DISCUSSION

A. ENVIRONMENT SETUP

To evaluate the performance of the proposed *HARNES* framework, we developed a simulation environment based on a structured grid network that represents urban road segments. The grid is implemented in MATLAB, and dynamic vehicle mobility is introduced using trace files generated from the Simulation of Urban Mobility (SUMO) [34]. These mobility traces are imported into MATLAB to simulate real-time changes in vehicle positions and connectivity. Thus, although the topology is regular, the vehicle movement and connectivity conditions remain dynamic and time-varying [34].

The road topology is modeled as a $2D$ grid graph, where each node represents a road intersection and is connected to up to 8 neighbors (vertical, horizontal, and diagonal), forming an 8-connected network. The number of nodes n ranges from 10 to 100, increasing in steps of 10. The grid dimensions are chosen to be approximately square by computing the number of rows and columns as:

$$\text{rows} = \lfloor \sqrt{n} \rfloor, \quad \text{cols} = \lceil n/\text{rows} \rceil \tag{7}$$

In each simulation trial, the first node (top-left corner) is set as the source, and the last node (bottom-right corner) is designated as the destination, enforcing end-to-end routing across the grid. Here, n denotes the nominal node count; since the grid is rounded up to the nearest rectangular rows $\times$ cols layout, the actual number of nodes is rows $\times$ cols $\geq n$, and the destination is the last node of that grid. To enable hierarchical routing, the grid is divided into four clusters using a 2×2 partitioning scheme. Each node is assigned a cluster ID based on its row and column location relative to the grid midpoint. A cluster head is selected for each region to facilitate intra-cluster coordination.

The control logic is managed by SDN controllers embedded in RSU at the cluster level. These RSU-embedded SDN controllers perform local routing decisions, monitor vehicle mobility, and facilitate inter-cluster communication. This architectural design ensures adaptive and low-latency decision-making, aligning with the SDN paradigm while preserving scalability [18]. Although this study does not yet use irregular real-world road maps, the adopted setup still provides a dynamic and controlled environment for evaluating clustering, controller coordination, and heuristic route discovery [18]. Future validation can be extended using SUMO-compatible real-world road-network sources, such as OpenStreetMap, which provide a practical basis for importing realistic urban layouts into mobility simulation environments [35].

TABLE 2. Simulation parameters setup.

Parameter	Value / Description
Node Count Range	10 to 100 (in steps of 10)
Road Network	Grid-based urban topology
Connectivity	8-connected (horizontal, vertical, and diagonal links)
Vehicle Dynamics	SUMO-generated mobility traces imported into MATLAB
Grid Dimensions	rows $= \lfloor \sqrt{n} \rfloor$, cols $= \lceil n/\text{rows} \rceil$
Start Node	Node 1 (top-left corner of the grid)
Goal Node	Last node, rows $\times$ cols (bottom-right corner of the grid)
Clustering Strategy	2×2 quadrant-based clustering
SDN Controller Location	Embedded at RSU level for each cluster
Algorithms Evaluated	EG-Dijkstra, A^*, Cluster-based Theta*, Cluster-based D*-Lite
Cost Metric	Euclidean distance
Path Length Metric	Number of hops
Computation Time Model	Based on expanded nodes and discovery messages
Simulation Tools	SUMO (mobility) and MATLAB (routing and analysis)

The simulation environment uses a consistent set of parameters across all experiments, including node count, connectivity, clustering strategy, and mobility dynamics. A summary of these parameters is presented in Table 2. The performance of each routing algorithm is evaluated using the following metrics:

- **Computation Time (CT):** Modeled search time in milliseconds, estimated from exploration effort scaled by algorithm complexity.
- **Discovery Messages (DM):** Total number of neighbor evaluations performed (search overhead).
- **Path Length (PL):** Number of hops from source to destination in the final computed path.
- **Expanded Nodes (EX):** Number of nodes visited or explored during the search.
- **Average Degree of Expanded Nodes (AD):** Mean number of neighbors for the expanded nodes.
- **Path Optimality Ratio (POR):** Ratio of the optimal path length obtained by EG-Dijkstra to the path length computed by a given algorithm.

B. PERFORMANCE COMPARISON

This subsection presents a comparative analysis of the evaluated routing algorithms across the considered performance metrics, including EG-Dijkstra, A^*, cluster-based $Theta^*$, cluster-based D^*-Lite, and the original $Theta^*$ and D^*-Lite

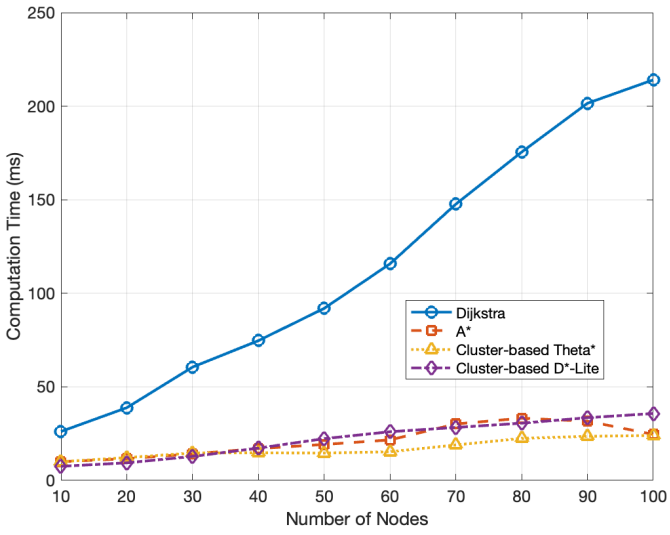


FIGURE 3. Computation Time comparison across different node counts. Cluster-based *Theta** consistently achieves the lowest time.

variants. Each metric captures a specific aspect of routing effectiveness or efficiency in a dynamic, grid-based vehicular network. The results are discussed below.

1) COMPUTATION TIME (CT)

Figure 3 illustrates the computation time required by each routing algorithm across varying node densities. It is evident that cluster-based *Theta** consistently outperforms the other algorithms in terms of computational efficiency. This can be attributed to two main factors: its *LOS* optimization, which reduces unnecessary expansions, and its cluster-based search scope, which limits the search to smaller local regions [36].

As the number of nodes increases from 10 to 100, EG-Dijkstra's algorithm shows a sharp linear increase, reaching over 220 ms at 100 nodes. This behavior is expected, as EG-Dijkstra explores the entire graph without any heuristics or spatial filtering. A^*, while more efficient than EG-Dijkstra due to its heuristic-based pruning, still incurs higher computational time than cluster-based algorithms, especially at lower densities, where its heuristic advantage is less pronounced.

Cluster-based D^*-Lite performs comparably to A^*, slightly surpassing it in larger grids. However, it does not leverage geometric optimizations such as *LOS*, which limits its performance gains during early-stage expansions.

In contrast, cluster-based *Theta** maintains the lowest computation time across all scales, staying below 30 ms even at 100 nodes. Its integration of clustering to reduce the search space) and *LOS* checks (to minimize path redirections) enables it to compute paths with fewer expanded nodes and lower discovery overhead, thereby significantly reducing time complexity [37]. Cluster-based *Theta** achieves up to a 42% reduction in computation time compared to A^*.

2) DISCOVERY MESSAGES (DM)

Figure 4 presents the number of discovery messages (i.e., neighbor evaluations) generated by each routing protocol across varying network sizes. This metric quantifies the control overhead during route computation and is especially critical in bandwidth-constrained or latency-sensitive vehicular environments [38].

As expected, EG-Dijkstra incurs the highest overhead, with discovery messages increasing rapidly as node density grows. Since EG-Dijkstra performs exhaustive graph traversal without any heuristic guidance, it leads to numerous neighbor checks—exceeding 650 messages at 100 nodes. This makes EG-Dijkstra inefficient for large-scale or real-time applications.

Cluster-based D^*-Lite shows a moderate increase in overhead with increasing node count. While its cluster-based design limits the search to specific regions, it does not incorporate heuristic prioritization or geometric path optimization. This leads to slightly more frequent neighbor evaluations, particularly at inter-cluster boundaries.

On the other hand, A^* significantly reduces the number of discovery messages compared to both EG-Dijkstra and D^*-Lite. Its Euclidean heuristic helps focus the search on promising paths, thereby reducing the messaging burden, especially at lower node counts. However, as the network grows, A^* still incurs moderate overhead.

Among all protocols evaluated, the cluster-based *Theta** achieves the lowest discovery message overhead in all scenarios. This is primarily due to its combination of heuristic directionality, line-of-sight optimization, and cluster-local planning. The protocol maintains overhead consistently below 80 messages, even as the node count reaches 100.

In conclusion, cluster-based *Theta** emerges as the most efficient algorithm in terms of communication overhead, highlighting its suitability for dynamic, *SDN*-managed vehicular networks where control traffic must be minimized. The overhead is reduced by up to 88% with cluster-based *Theta** compared to EG-Dijkstra.

3) PATH LENGTH (PL)

Figure 5 illustrates the average path length computed by each routing algorithm across varying node densities. Path length here refers to the number of hops between the source and the destination and is a core metric for evaluating routing efficiency [39].

The results show that all four protocols, i.e., EG-Dijkstra, A^*, cluster-based *Theta**, and cluster-based D^*-Lite, achieve almost identical path lengths across all node counts. This is expected, as the network is grid-structured and the destination remains fixed relative to the source, resulting in predictable, similarly optimal routes.

However, cluster-based *Theta** consistently produces paths that are marginally shorter than the others. This improvement, though small, is meaningful and can be attributed to its *LOS* capability, which allows the algorithm to skip intermediate hops when direct paths are available within or across clusters.

In conclusion, cluster-based *Theta** delivers the best path efficiency, albeit with marginal gains. Its ability to integrate geometric path shortcuts without sacrificing optimality positions it as a well-balanced routing choice.

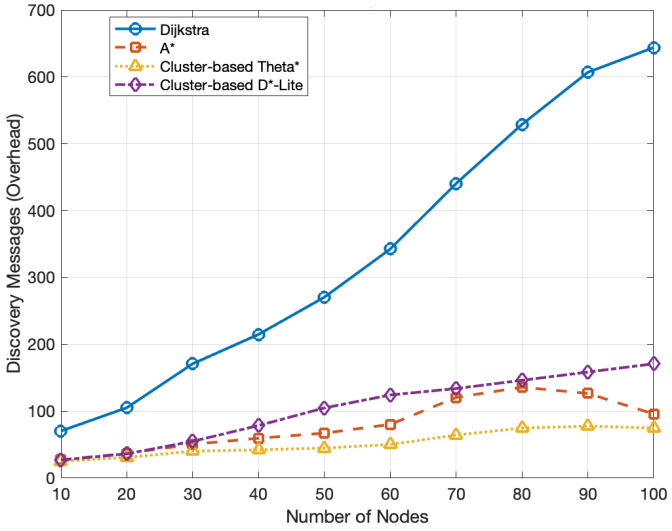


FIGURE 4. Discovery messages (overhead) comparison. Cluster-based *Theta** consistently incurs the lowest control overhead.

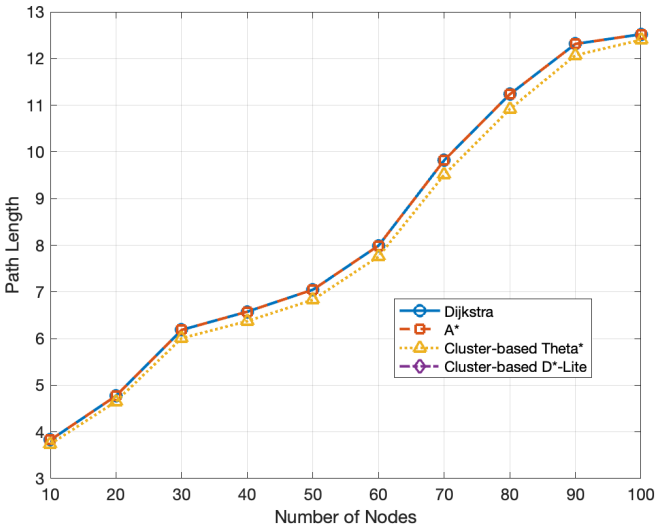


FIGURE 5. Path length comparison across routing protocols. Cluster-based *Theta** shows marginally shorter paths due to LOS optimization.

4) EXPANDED NODES (EX)

Figure 6 shows the number of nodes expanded during route discovery for each routing algorithm across increasing node densities. This metric reflects the extent of the search space traversed by the algorithm and directly correlates with computational effort and memory usage [40].

As expected, EG-Dijkstra exhibits the highest number of expanded nodes. With no heuristic or spatial constraints, it exhaustively explores the graph, resulting in a steep increase in node expansions that surpasses 90 at 100 nodes.

A^* significantly reduces the number of expanded nodes compared to EG-Dijkstra by using an Euclidean distance heuristic to guide the search toward the goal. As a result, it expands fewer irrelevant nodes and maintains expansion counts below 20 across all scenarios.

Cluster-based D^*-Lite further limits the expansion region by restricting the search within localized clusters. While it

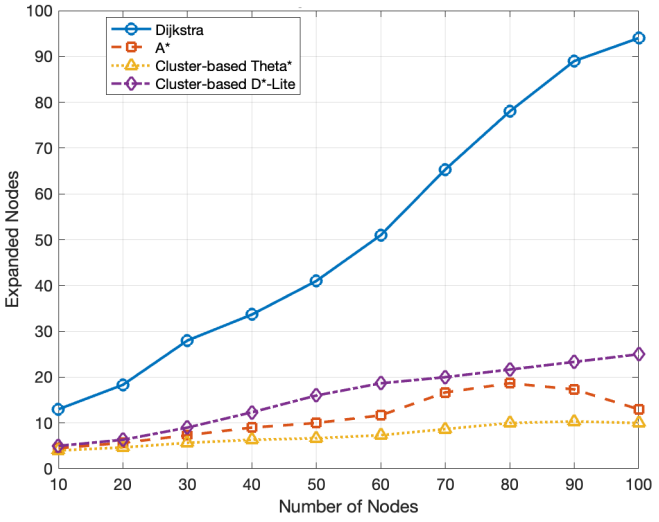


FIGURE 6. Expanded nodes comparison across routing protocols. Cluster-based *Theta** maintains the lowest expansion across all node counts.

does not use a heuristic for prioritization, its hierarchical structure helps reduce the number of nodes explored, especially in denser scenarios.

Cluster-based *Theta** achieves the lowest expansion count among all evaluated algorithms. Its performance is driven by two key factors: the clustering mechanism, which limits the search scope, and its ability to leverage *LOS* paths, thereby avoiding unnecessary expansions of intermediate nodes. It consistently expands fewer than 11 nodes, regardless of network size.

In summary, cluster-based *Theta** demonstrates superior search efficiency by expanding the fewest number of nodes. This characteristic makes it highly scalable and well-suited for time-sensitive and resource-constrained vehicular environments. Cluster-based *Theta** limits exploration, reducing the number of expanded nodes by 47% compared to EG-Dijkstra.

5) AVERAGE DEGREE OF EXPANDED NODES (AD)

Figure 7 illustrates the average degree of expanded nodes for all routing protocols across varying node densities. This metric reflects the structural richness or connectivity of the nodes selected during path computation. A higher degree may indicate that the algorithm is expanding nodes located in densely connected regions, which can influence both flexibility and computational complexity [41].

As shown in the figure, A^* has the highest average degree. This suggests that A^* tends to expand nodes that are more centrally located in the graph or more connected to neighboring nodes. While this can provide more routing options, it also increases the branching factor, which may lead to additional computational burden.

EG-Dijkstra shows a steadily increasing trend in average degree, though it remains lower than A^* in most scenarios. This behavior aligns with its exhaustive nature: node expansion follows a breadth-first search without prioritization, resulting in moderate connectivity within the explored region.

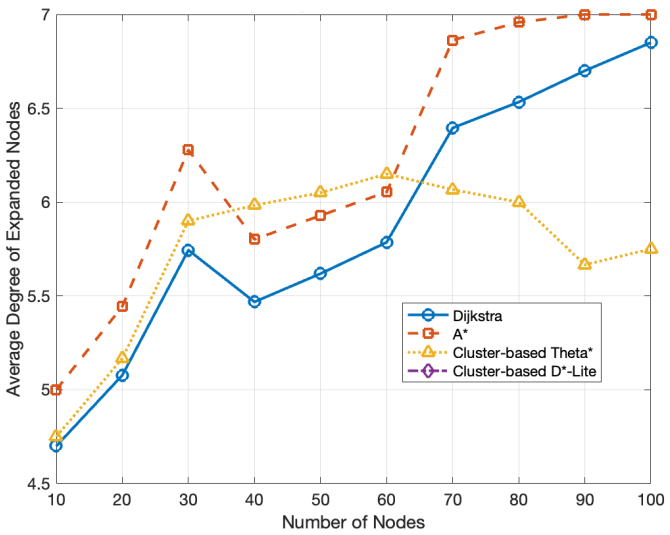


FIGURE 7. Average degree of expanded nodes. A sharp dip in Cluster-based *Theta** appears around 60 nodes due to cluster boundary effects.

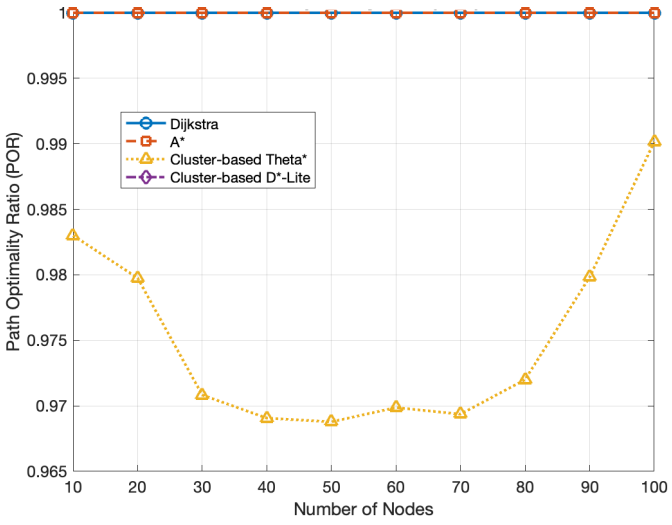


FIGURE 8. Path Optimality Ratio (POR) across varying node densities. Cluster-based *Theta** shows slightly lower POR at mid-scale densities due to LOS-based shortcuts.

Cluster-based *Theta** maintains a relatively stable average degree across all node counts, hovering around 6. However, a noticeable dip appears around 60 nodes. This drop is likely due to the temporary spatial arrangement of clusters and the expansion of nodes near boundaries or less-connected areas. At this point, the grid structure and cluster assignment may lead to more expansions near the periphery or in transitional zones with fewer neighbors, thereby reducing the average degree. The metric stabilizes again as node count increases and the topology becomes denser.

Cluster-based D^*-Lite remains mostly flat in the plot and close to the lower bound. Since it operates within tightly bounded clusters and does not explore far beyond local connections, the expanded nodes are often near the periphery or have fewer neighbors, resulting in a lower average degree.

In summary, while A^* explores high-connectivity regions, potentially increasing cost, cluster-based *Theta** achieves an ideal balance by expanding moderately connected nodes with spatial awareness. Its slight fluctuations reflect its adaptability to dynamic spatial conditions in clustered topologies.

6) PATH OPTIMALITY RATIO (POR)

Figure 8 presents the *POR* of the evaluated routing protocols. In this study, *POR* is defined as the ratio of the optimal path length obtained by EG-Dijkstra to the path length computed by a given routing algorithm in terms of hop count [42], [43]. A value of 1.0 indicates a fully optimal path, while values slightly below 1.0 indicate a small deviation from the minimum-hop route.

As expected, both EG-Dijkstra and A^* maintain a perfect *POR* of 1.0 across all node densities, confirming that their path decisions are consistently optimal. Similarly, cluster-based D^*-Lite also achieves a *POR* of 1.0, indicating that its localized path planning does not compromise optimality in hop count, consistent with its design, which resembles cluster-constrained EG-Dijkstra.

TABLE 3. Comparison of original and cluster-based *Theta** and D^*-Lite averaged over node densities from 10 to 100.

Method	Avg. CT (ms)	Avg. DM	Avg. EX
Original *Theta**	16.39	54.5	7.6
Cluster-based *Theta**	14.95	53.7	6.7
Original D^*-Lite	23.05	77.8	11.0
Cluster-based D^*-Lite	16.05	56.1	7.0

In contrast, cluster-based *Theta** exhibits *POR* values slightly below 1.0, especially between 30 and 70 nodes. This behavior reflects that *Theta** prioritizes line-of-sight-guided geometric efficiency over strict hop-count minimization. As a result, the selected route may occasionally be slightly longer than the minimum-hop path obtained by EG-Dijkstra, even though it remains highly efficient in practice. This deviation is small, however, and stays within about 3.1%, indicating that the proposed method preserves near-optimal path quality while benefiting from lower routing overhead.

As the node density increases beyond 80, the *POR* of cluster-based *Theta** moves closer to 1.0. This suggests that in denser topologies, the line-of-sight-guided paths align more closely with minimum-hop routes, improving the balance between geometric efficiency and hop-based optimality.

Overall, although cluster-based *Theta** trades off a small amount of hop-based optimality, it compensates with reduced computational overhead and message cost, as shown in the previous metrics, making it a strong candidate for real-time, resource-constrained applications.

7) COMPARISON WITH ORIGINAL *THETA** AND D^*-LITE

To further justify the benefit of the proposed cluster-based design, we compared cluster-based *Theta** and cluster-based D^*-Lite with their original counterparts. Table 3 reports the average performance over node densities from 10 to 100 using three key metrics: Computation Time (CT), Discovery Messages (DM), and Expanded Nodes (EX).

The results show that the proposed cluster-based variants consistently reduce routing overhead. In particular,

cluster-based *Theta** achieves 8.79% lower computation time, 1.47% fewer discovery messages, and 11.84% fewer expanded nodes compared to the original *Theta**. The gains are even more significant for cluster-based *D**-Lite, which achieves 30.37% lower computation time, 27.89% fewer discovery messages, and 36.36% fewer expanded nodes relative to the original *D**-Lite.

These results confirm that the proposed clustering mechanism effectively narrows the search scope and improves routing efficiency. The improvement is more pronounced for *D**-Lite, whereas for *Theta** the gain remains moderate because the original algorithm already benefits from strong heuristic and LOS guidance.

V. CONCLUSION

This work presented *HARNES*, a heuristic routing framework that integrates clustering strategies with SDN-enabled coordination to improve scalability and efficiency in dynamic vehicular networks. We developed cluster-based extensions of *Theta** and *D**-Lite, which leverage localized search and *LOS* optimization to reduce routing overhead while preserving path quality. The overall results confirm that *HARNES*, particularly the cluster-based *Theta**, provides an effective and SDN-compatible solution for intelligent routing in large-scale vehicular environments. The additional comparison with the original *Theta** and *D**-Lite further supports the benefit of the proposed cluster-based design. Although minor trade-offs in hop count were observed under dense topologies, the framework maintained strong scalability and responsiveness through cluster-aware expansion control. Future work will extend this study to validate and adapt real-world testbeds for heterogeneous vehicular and 5G/6G-enabled communication scenarios. It will also include direct benchmarking against closely related SDN- and cluster-based routing schemes, particularly SE-CEGRS, under harmonized evaluation settings. Although the proposed framework improves routing efficiency, it incurs additional coordination cost associated with cluster formation and cluster-head selection, which should be further analyzed under highly dynamic re-clustering scenarios.

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